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Performance of State-of-the-Art Machine Learning Methods on Out-of-Distribution Data in Tabular Regression: A Survey

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Abstract

$$P^{train} \neq P^{test}$$

1 Introduction

Artificial Intelligence (AI) [1] is a broad computer science field that studies about softwares that can mimic the human’s capability of learning and problem solving. Machine Learning (ML) is a sub-field of AI, which [2] applies statistical methods to explore the underlying correlation among the features and the target. Nonetheless, ML faces difficulties when dealing with high data volume and dimensions. A sub-field of ML has been studied to overcome these challenges, i.e. Deep Learning [3], which has proven its superiority over conventional ML approaches, and yielded prospective results in Natural Language Processing (NLP), Computer Vision (CV), Large Language Models (LLMs), and also being applied to other industries ([4], [5]), e.g. healthcare, finance, energy, and agriculture.

Since 1940s [6], several ML/DL models have been proposed, and various benchmarks have been introduced to compare the performance among them. A critical fault is that these benchmarks often ([7], [8], [9], [10]) overlook the distributional shifts between the training data and deployment data, alternatively, they assume that the traing and testing sets are independent and identically distributed (i.i.d). These inherent distributional shifts are due to various factors, e.g. [8] sample biases, environment changing, data generation errors, and [11] learning spurious correlations. Many studies have been contributed to showing [11] distributional shifts in real data are roadblock to the ML/DL applications to other fields, and many [12] benchmarks and [13] AI practioners failed to analyse the models’ performance on Out-of-Distribution (OOD) data. In Computer Vision, [14] showed that recent image classifiers fail to OOD generalization and [15] introduced a benchmark for performance evaluation among object detectors, namely ObjectNet, which showed performance drop of top models, from 2012 to 2018, in comparision to other standard benchmarks, e.g. ImageNet. Regarding Natural Language Processing, [16] proposed a benchmark, called CheckList, that leads to failure in semantic analysis due to tiny variations in testing texts compared to those in the training set.

Although ([17], [18]) tabular data is the primary data structure to retain data for training models in numerous areas, e.g. finance ([19], [20]), healthcare ([21], [22]) and manufacturing sectors [23], the number of standardized tabular regression benchmarks are very limited ([24], [17]). Therefore, this report serves as a benchmark consisting of various train/test splitting strategies on real datasets to evaluate model’s capability towards OOD generalization on tabular regression tasks. The key contributions of our work are:

- Introduction to a comprehensive benchamrk consisting of 26 real-world, low dimensional (up to 20 features) tabular datasets specifically for regression tasks. These datasets are subsequently partitioned into several train/test pairs aiming for evaluating the models’ OOD generalization. To our knowledge, this marks the first benchamrk that encompasses certain approaches to split datasets for OOD extrapolation assessment.
- Conduct an experimental study to quantify and rank the robustness of a broad spectrum of standard ML/DL models on unseen distributions.

- Give insightful analysis to explore what characteristics of the proposed ML/DL methods impact their performance when generalising to unseen data regions.

Before proceeding, it is recommended to see the Appendices for the use of Abbreviations and Mathematics Notations within this report.

2 Related works

The aim of this report is to investigate the performance of widely used ML models on Out-of-Distribution (OOD) tabular data. Thus, plenty of papers were introduced to formally define what it means for models to extrapolate on unseen data. Many papers ([25], [26], [27]) have supported that extrapolation to OOD data refers to models predicting data points lying beyond the **convex hull of training samples**. Nevertheless, a more prevalent definition of OOD data highlights the **distributional shifts between the training and testing sets**, and is commonly used to compare models' generalization on OOD data ([9], [7], [12], [24], [11], [17]). To our knowledge, there are none of benchmarks for tabular regression tasks on OOD data that is built upon the first definition. Meanwhile, there are contributions, though still very limited, to benchmarking models on OOD data, based on distributional shifts, in tabular regression tasks, e.g. **Wild-Tab** [17], **Shifts Dataset** [7], and an unnamed benchmark provided by [24].

The **Wild-Tab** benchmark [17] evaluates models' robustness under covariate, subpopulation, and hybrid shifts of large-sized real-world datasets. This report also conducted an empirical study by implementing baseline DL models, e.g. MLP-PLR, FT-Transformer, and 10 other OOD generalization techniques, e.g. IRM, ERM, and GroupDRO, that are specifically designed for tabular regression tasks, and gave certain insightful analysis, i.e. ERM outperforms other models, and the impact of model architecture and hyperparameter tuning methods on models' performance. In contrast, [24] offers a benchmark to compare between the tree-based models and DL models on OOD data in tabular regression tasks. More specifically, [24] leverages 45 tabular datasets, mostly from OpenML [28], and cleans and prepares properly. Later, they benchmarked 3 tree-based models, i.e. Random Forest, XGBoost, and Gradient Boosting Trees (or HistGradient Boosting Trees in case of categorical inputs) and 4 distinct DL models, i.e. classical MLP, ResNet, FT-Transformer, and SAINT. Also, they clearly interpreted why tree-based models outperform DL models in tabular regression on OOD data regardless of models' complexity. Despite limited available benchmarks for tabular regression on OOD data, these prior works inspired us to conduct a more comprehensive empirical study to compare the performance of various ML/DL models on OOD data that are created by applying diverse splitting strategies to real-world datasets.

3 Problem settings

3.1 Regression formulation

Throughout this report, we are dealing with tabular regression tasks, let us first formally recall the formulation of this problem. To begin with, we are given an input space or domain, denoted as $\mathcal{X}$, the objective is to [29] induce an approximation function, denoted as $\hat{f} : \mathcal{X} \mapsto \mathcal{Y} \subseteq \mathbb{R}$, such that, $\hat{f} \approx f$, where f is the true function representing the

relationship $\mathcal{X} \mapsto \mathcal{Y}$. To quantify how well the predictor/model $\hat{f}$ performs, several metrics have been proposed, typically Mean Squared Error (MSE), Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE). In practice, these evaluation metrics are leveraged as objective functions to train the models. Formally, the models repeatedly update their parameters to [11] diminish the population risk.

Definition 1. (True) Population Risk [8]

$$\mathcal{R}_{\text{true}}(\hat{f}) \equiv \mathcal{R}_{\mathbf{x} \in \mathcal{X}}(\hat{f}) \stackrel{\text{def}}{=} \mathbb{E}_{\mathbf{x} \in \mathcal{X}}[\ell(\hat{f}(\mathbf{x}), f(\mathbf{x}))]$$

Nonetheless, in real-world scenarios, it is almost impossible to capture all behaviours of $\mathcal{X}$ to train the models, e.g. self-driving car systems [11], dog/cat classification [8], and disease diagnosis for patients in all hospitals [10]. Alternatively, the model $\hat{f}$ is trained on a **sub-domain** $\mathcal{X}^{\text{tr}} \subset \mathcal{X}$, and hence, the model $\hat{f}$ is trying to optimize the sample risk or empirical risk, not the population risk as expected. In other words, while training on $\mathcal{X}^{\text{tr}} \subset \mathcal{X}$, the predictor $\hat{f}$ is expected to perform on $\mathcal{X}^{\text{te}} \subset \mathcal{X}$ as well as it does on $\mathcal{X}^{\text{tr}}$.

Definition 2. Empirical Risk [8]

$$\mathcal{R}_{\text{emp}}(\hat{f}) \equiv \mathcal{R}_{\mathbf{x} \in \mathcal{X}^{\text{tr}}}(\hat{f}) \stackrel{\text{def}}{=} \mathbb{E}_{\mathbf{x} \in \mathcal{X}^{\text{tr}}}[\ell(\hat{f}(\mathbf{x}), f(\mathbf{x}))]$$

Despite the impossibility to gather the whole domain $\mathcal{X}$, practioners and researchers have endeavoured to collect samples to form the training set $\mathcal{X}^{\text{tr}}$ that behaves somewhat similar to the whole population or domain $\mathcal{X}$. Thereby, the model $\hat{f}$ is expected to [8] make the **Empirical Risk** $\mathcal{R}_{\text{emp}}$ approximates the **True Risk** $\mathcal{R}_{\text{true}}$. Formally, this is

$$\arg \min_{\hat{f}} \mathcal{R}_{\text{emp}}(\hat{f}) \approx \arg \min_{\hat{f}} \mathcal{R}_{\text{true}}(\hat{f})$$

To measure the similarity between the sample training set $\mathcal{X}^{\text{tr}}$ and the true population $\mathcal{X}$, we formally present two commonly used concepts, i.e. **distributional shifts** and **convex hull of data** as below.

3.2 Distributional shifts

Throughout this section, we aim to estimate the similarity between $\mathcal{X}^{\text{tr}}$ and $\mathcal{X}^{\text{te}}$ based on their statistical properties, more precisely, their **joint distributions**, i.e. $\mathbf{P}^{\text{tr}}(\mathbf{X}, \mathbf{y})$ and $\mathbf{P}^{\text{te}}(\mathbf{X}, \mathbf{y})$. More detailed, the **marginal distribution** of $\mathcal{X}^{\text{tr}}$ and $\mathcal{X}^{\text{te}}$ are written as $\mathbf{P}^{\text{tr}}(\mathbf{X})$ and $\mathbf{P}^{\text{te}}(\mathbf{X})$, respectively. On the other hand, the **conditional distribution** interprets distribution of $\mathcal{Y}$ with regards to $\mathcal{X} \mapsto \mathcal{Y}$, written as $\mathbf{P}(\mathbf{y} \mid \mathbf{X})$.

A pervasive hypothesis ([7], [8]) while training and validating models is that the train and test sets are independent and identically distributed (i.i.d). In other words, the test data $\mathcal{X}^{\text{te}}$ are considered statistically unchanged in comparison with the train data $\mathcal{X}^{\text{tr}}$, and this assumption has vastly contributed to the development of ML/DL models in the past decades [30].

Definition 3. No shifts - In-Distribution (ID) [8]

The test data $\mathcal{X}^{\text{te}}$ is considered to be In-Distribution (ID), or not shifted, when the following holds:

- $\mathbf{P}^{\text{te}}(\mathbf{X}) = \mathbf{P}^{\text{tr}}(\mathbf{X})$
- $\mathbf{P}^{\text{te}}(\mathbf{y} \mid \mathbf{X}) = \mathbf{P}^{\text{tr}}(\mathbf{y} \mid \mathbf{X})$

As a result, the models $\hat{f}$ show excellence performance on validation samples, but are vulnerable to real-world data. Therefore, it is imperative to detect and address these issues, so that the models can perform better on real data. In practice, there are [8] two primary types of distributional shifts, which are shifts in feature space $\mathcal{X}$, known as **covariate shifts** and shifts in label space $\mathcal{Y}$, known as **concept or semantic shift**. Both of them contribute to the fact that $\mathbf{P}^{\text{tr}}(\mathbf{X}, \mathbf{y}) \neq \mathbf{P}^{\text{te}}(\mathbf{X}, \mathbf{y})$, such test data are said to be **Out-of-Distribution (OOD)**, and thus it downgrades the performance of models $\hat{f}$. Recall our initial objective is to induce an approximation function $\hat{f} : \mathcal{X} \mapsto \mathcal{Y}$, the mapping direction is essential to construct the upcoming definitions.

Definition 4. Covariate shifts [31]

Covariate shift is defined when the following holds:

- $\mathbf{P}^{\text{te}}(\mathbf{X}) \neq \mathbf{P}^{\text{tr}}(\mathbf{X})$
- $\mathbf{P}^{\text{te}}(\mathbf{y} \mid \mathbf{X}) = \mathbf{P}^{\text{tr}}(\mathbf{y} \mid \mathbf{X})$

Definition 5. Concept/Semantic shifts [31]

Concept shift is defined when the following holds:

- $\mathbf{P}^{\text{te}}(\mathbf{X}) = \mathbf{P}^{\text{tr}}(\mathbf{X})$
- $\mathbf{P}^{\text{te}}(\mathbf{y} \mid \mathbf{X}) \neq \mathbf{P}^{\text{tr}}(\mathbf{y} \mid \mathbf{X})$

Besides these dataset shifts, [31] theoretically there is another situation that possibly occur in real-world scenario, i.e.

- $\mathbf{P}^{\text{te}}(\mathbf{X}) \neq \mathbf{P}^{\text{tr}}(\mathbf{X})$
- $\mathbf{P}^{\text{te}}(\mathbf{y} \mid \mathbf{X}) \neq \mathbf{P}^{\text{tr}}(\mathbf{y} \mid \mathbf{X})$

3.3 Convex hull of data

Another definition, though rarely used, originated from a geometric view that leverages the definition of **Convex Hull**.

Definition 6. Convex Hull [32] [33]

Given a set of m points lying in n -dimension space, $\mathcal{S} \triangleq \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m\} \subset \mathbb{R}^n$. The Convex Hull induced by $\mathcal{S}$ is the smallest convex set that completely contains $\forall \mathbf{x} \in \mathcal{S}$. Formally:

$$\text{Conv}\{\mathcal{S}\} \stackrel{\text{def}}{=} \left\{ \mathbf{v} = \sum_{i=1}^m \lambda_i \mathbf{x}_i \mid \forall \lambda_i \geq 0 \wedge \sum_{i=1}^m \lambda_i = 1 \right\}$$

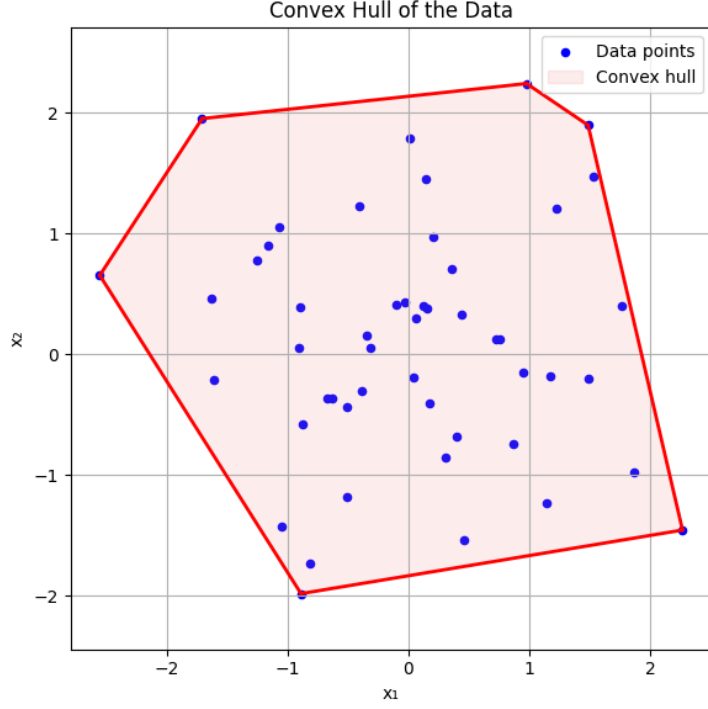


Figure 1: Convex Hull of randomly generated points in 2D visualization

Following the above definition, the OOD generalization can be interpreted as extrapolation.

Definition 7. Interpolation vs. Extrapolation with Convex Hull [25]

Given a model $\hat{f}$ trained on a set of points $\mathcal{X}^{\text{tr}}$, the convex hull of $\mathcal{X}^{\text{tr}}$, denoted as $\text{Conv}(\mathcal{X}^{\text{tr}})$. For any point $\mathbf{x} \sim \mathcal{X}^{\text{te}}$ and $\mathcal{X}^{\text{te}} \cap \mathcal{X}^{\text{tr}} = \emptyset$:

- Interpolation occurs when $\mathbf{x} \in \text{Conv}(\mathcal{X}^{\text{tr}})$.
- Extrapolation occurs when $\mathbf{x} \notin \text{Conv}(\mathcal{X}^{\text{tr}})$.

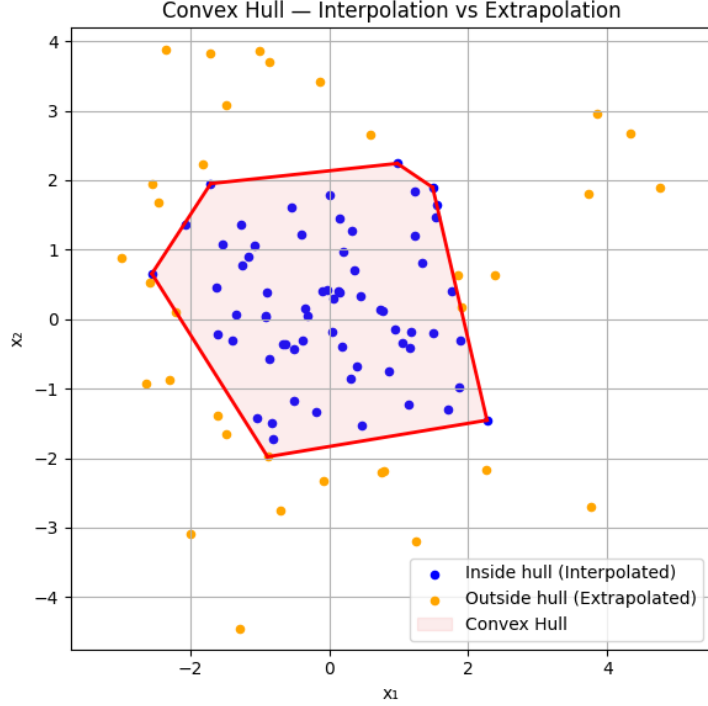


Figure 2: Interpolation vs. Extrapolation with Convex Hull in 2D visualization

In 2D space, the boundary between interpolation and extrapolation can be clearly distinguished. Nonetheless, in higher dimensional space, it is highly vulnerable to the effect of the Curse of Dimensionality [34]. More specifically, [35] the distances among the data points become more scattered, and the commonly used distance metrics, e.g. Euclidean-based, may become invalid, owing to the phenomenon of distance concentration. A (mis)conception emerged that higher precision on training set implies better models’ generalization/extrapolation on unseen data regions, which was proven to be completely false, ([25], [35]), this issue is known as overfitting, which may be lessened by the expansion of number of training samples ([25], [35]). However, the datasets used within this report is low-dimensional data, at most 20 features, and the training size ranging from tens to hundreds of thousands samples, which may mitigate the impact of the Curse of Dimensionality.

4 Benchmark

4.1 Dataset

This section offers an overview of the dataset used and the sources of them. In addition, some of the datasets below are different from the original datasets due to sample and features removals to suit the need of this report. Moreover, the datasets were modified so that the target column is the last column, from left to right, in tabular format.

Table 1: Dataset overview

Original name	Dataset name	#samples	#features
218_house_8L	218_house_8L	22,784	8
House-16H	house	22,784	16
California Housing	california	20,640	8
Elevators	elevators	16,599	18
Physicochemical Properties of Protein Tertiary Structure	CASP	45,730	9
Gas Turbine CO and NOx Emission Data Set	gas_turbine_co_and_nox_emission	36,733	10
Condition Based Maintenance of Naval Propulsion Plants	CBM_1	11,934	16
Condition Based Maintenance of Naval Propulsion Plants	CBM_2	11,934	16
Zurich public transport delay data	delays_zurich_transport	800,000	8
diamonds	diamonds	53,940	6
nyc-taxi-green-dec-2016	nyc_taxi_green_dec_2016	581,835	9
house_sales	house_sales	21,613	15
medical_charges	medical_charges	163,065	3
MiamiHousing2016	MiamiHousing2016	13,932	13
sulfur	sulfur_1	10,081	5
sulfur	sulfur_2	10,081	5
Goodreads-books	books	10,348	3
FIFA 22 complete player dataset	players_22	17,041	10
Taxi Duration for Domain Generalization	taxi	800,000	13
UCI Bike Sharing for Domain Generalization	bike	17,379	5
Vessel Power Estimation	power_consumption	567,442	9
3droad	3droad	434,874	3
keggdirected	keggdirected	48,827	20
kin40k	kin40k	40,000	8
protein	protein	45,730	9
tamielectric	tamielectric	45,781	3

4.2 Train/Test splitting strategy

Due to length limits, through out this report, the dataset [CASP](#) is utilized as an example to visualize how the splitting approaches influence the data distributions. The remaining statistics summary of both raw and splitted datasets are given in the GitHub repo of this report.

In addition, for extensive datasets, e.g. `taxi` and `delays_zurich_transport`, which is more than 1,000,000 samples, they are randomly selected 800,000 samples to guarantee the distributions of the original datasets. Besides, the train/test proportion used within this report is 70% and 30%, respectively. To guarantee reproducibility of this work, we chose a list of 10 random seeds, i.e. `SEEDS = [0, 42, 19, 2, 23, 11, 37, 29, 57, 5]`.

4.2.1 Random strategy

Throughout this report, to highlight the distinction among the common splitting strategy and the proposed strategies, and to show the difference between the models' performance on ID and OOD data, we leverage a `sklearn` built-in function, i.e. `train_test_split()`, to randomly sample the train and test sets, so as to construct iid train/test sets. More specifically, the distributions of train and test sets are identical, or $\mathbf{P}^{\text{tr}}(\mathbf{X}, \mathbf{y}) = \mathbf{P}^{\text{te}}(\mathbf{X}, \mathbf{y})$.

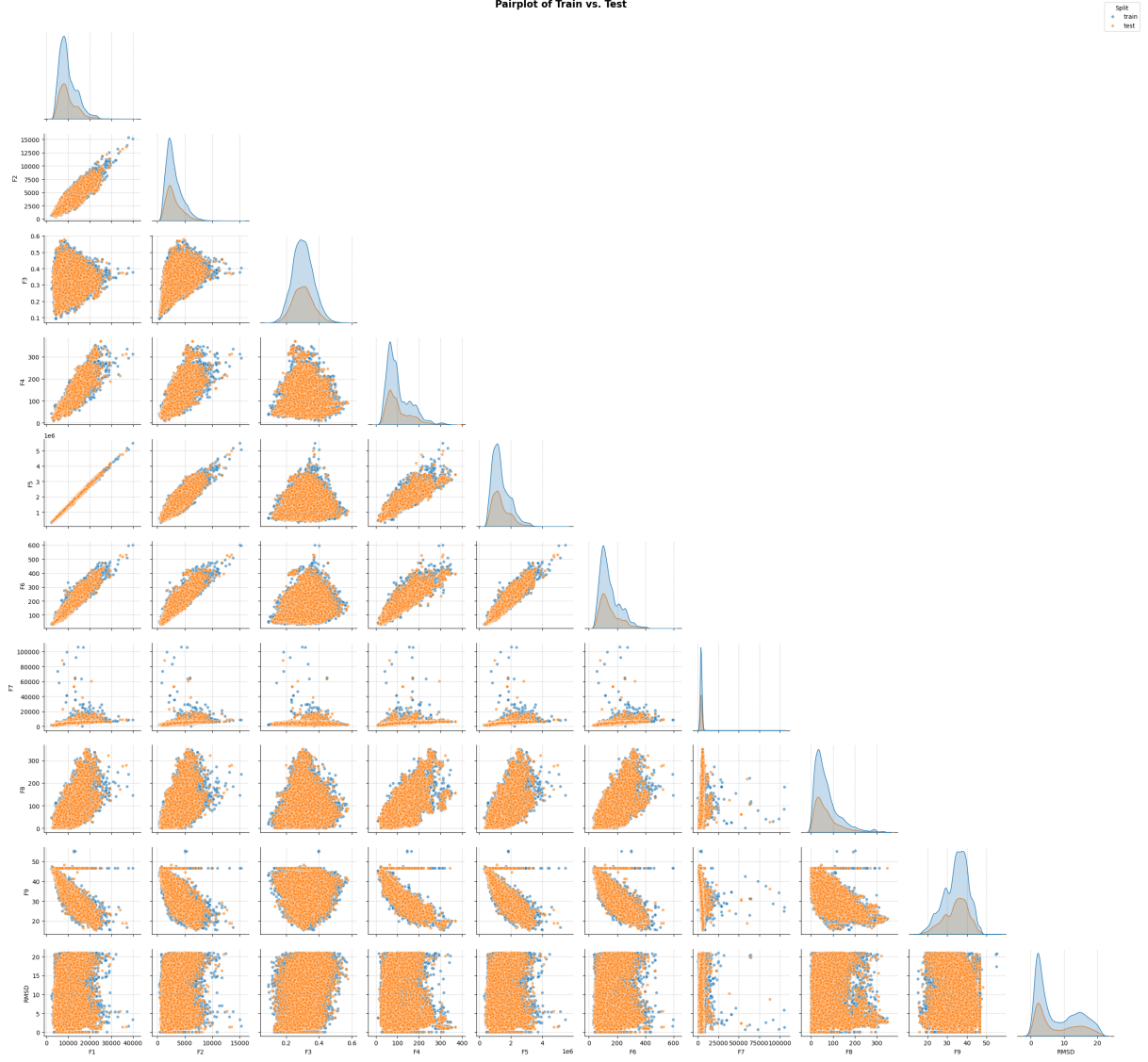


Figure 3: Pairplot of Train vs. Test

4.2.2 Distribution-based strategy

In this approach, we look at the distributions of each feature and the conditional distributions between the features and outputs to construct the OOD data. Recall **Definition 4. Covariate shifts** and **Definition 5. Concept shifts**, we would like to simulate the behaviour of these distributional shifts.

Regarding **Covariate shifts**, we proposed a fairly naive approach. More specifically, we rely on the distribution of each feature to construct the train/test sets, thereby, the number of such train/test set pairs depending how many features each dataset has.

Algorithm 1: MarginalDistributionSplit: Covariate Shift Splitting

Require: Dataset $\mathcal{D} = (\mathbf{X}, \mathbf{y})$ with features $\{\text{feat}_1, \dots, \text{feat}_m\}$ and target $\mathbf{y}$; test ratio $\tau \in (0, 1)$

Ensure: Multiple train/test splits saved as `train_idx.parquet` and `test_idx.parquet`

- 1: $idx \leftarrow 0$
- 2: **for all** $\text{feat} \in \{\text{feat}_1, \dots, \text{feat}_m\}$ **do**

```

3:    $mask \leftarrow \text{DISTRIBUTIONBASEDSELECTION}(\mathcal{D}, \text{feat}, \tau)$ 
4:   if  $\frac{\sum mask}{|\mathcal{D}|} < \tau$  then
5:       Print “Skip feature {feat} due to insufficient test samples”
6:       continue
7:   end if
8:    $X_{train} \leftarrow X[\neg mask]; y_{train} \leftarrow y[\neg mask]$ 
9:    $X_{test} \leftarrow X[mask]; y_{test} \leftarrow y[mask]$ 
10:   $\mathcal{D}_{train} \leftarrow \text{CONCATCOLUMNS}(X_{train}, y_{train})$ 
11:   $\mathcal{D}_{test} \leftarrow \text{CONCATCOLUMNS}(X_{test}, y_{test})$ 
12:  Save  $\mathcal{D}_{train}$  as train_idx.parquet
13:  Save  $\mathcal{D}_{test}$  as test_idx.parquet
14:   $idx \leftarrow idx + 1$ 
15: end for

```



Figure 4: Pairplot of Covariate_Shift train_0/test_0

In contrast, to mimic the behaviour of **Concept shifts**, we utilized the Metafeature-based

approach from [12]. Let us briefly summarize how this technique works.

Now, supposing a given dataset $\mathcal{D} = \{(\mathbf{x}_i, y_i) \mid \mathbf{x}_i \in \mathcal{X}, y_i \in \mathcal{Y}\}$. [36] A **meta-feature** $\mathcal{M} \ni M : \mathcal{D} \mapsto \mathbb{R}^k$, is defined as:

$$M(\mathcal{D}) = \sigma(m(\mathcal{D}, h_m), h_s)$$

where $\sigma(\cdot)$ is a characterization measure, $m(\cdot)$ is a summarization function, h_m and h_s are associated hyperparameters for $m(\cdot)$ and $\sigma(\cdot)$, respectively. We do not delve deeper into this definition as it is not our main goal in this report. However, it is imperative to know that the proposed **meta-feature** $M(\cdot)$ distils the dataset $\mathcal{D}$ into the set of k values that can characterize and predict the performance of the algorithms applied to $\mathcal{D}$.

Next, we are now given a labeled dataset $D = \{(\mathbf{x}_i, y_i) \mid \mathbf{x}_i \in \mathbf{X}, y_i \in \mathbf{y}\}$, and D is not representative of the whole population. The objective of [12] is to partition D into $D^{\text{tr}} \cup D^{\text{te}} = D$, where $D^{\text{tr}} \cap D^{\text{te}} = \emptyset$ and their meta-characteristics are different. Let $d \in D$ is a sample from D , [12] defines the following notions:

- Meta-feature extractor $M : d \mapsto \mathbb{R}^k$, that distils d into a vector of k meta-features values, defined as:

$$M(d) = [M_1(d), M_2(d), \dots, M_k(d)]$$

- For each meta-feature M_j , the directed distance is defined as:

$$\text{dist}_j(D^{\text{tr}}, D^{\text{te}}) = \begin{cases} \frac{M_j(D^{\text{tr}})}{M_j(D^{\text{te}})} & , \text{ if } M_j(D^{\text{tr}}) \cdot M_j(D^{\text{te}}) \neq 0 \\ 0 & , \text{ otherwise} \end{cases}$$

- A scalar imbalance measure is defined as:

$$\text{imb}(d) = \frac{\min_c n_c(d)}{\max_c n_c(d)}$$

where $n_c(\cdot)$ counts the label c in d .

- The imbalance objective is defined as:

$$o_{\text{imb}}(D^{\text{tr}}, D^{\text{te}}) = \left| \text{imb}(D^{\text{tr}}) - \text{imb}(D^{\text{te}}) + \lambda \left(1 - \min[\text{imb}(D^{\text{tr}}), \text{imb}(D^{\text{te}})] \right) \right|$$

where λ is the penalizing hyperparameter.

Eventually, [12] subjects to optimizing the multi-objective vectors:

$$\text{obj}(D^{\text{tr}}, D^{\text{te}}) = [\text{dist}_1(D^{\text{tr}}, D^{\text{te}}), \text{dist}_2(D^{\text{tr}}, D^{\text{te}}), \dots, \text{dist}_k(D^{\text{tr}}, D^{\text{te}}), o_{\text{imb}}(D^{\text{tr}}, D^{\text{te}})]$$

that minimizes the directed distance for each meta-feature while maximizing the balance between classes, by the utilization of the Evolutionary Algorithm.

In this report, we chose **mutual information** as our information-theoretic meta-feature. In brief, [12] **mutual information** measures the statistical correlation between covariates and target, and is computed as [37]:

$$\mathbf{I} = [I_1(\mathbf{X}_1; \mathbf{y}), I_2(\mathbf{X}_2; \mathbf{y}), \dots, I_m(\mathbf{X}_m; \mathbf{y})]$$

where:

$$I_i(\mathbf{X}_i; \mathbf{y}) = H(\mathbf{X}_i) + H(\mathbf{y}) - H(\mathbf{X}_i, \mathbf{y})$$

the Shanon's Entropy $H(\mathbf{X}_i)$, $H(\mathbf{y})$ are defined as:

$$H(\mathbf{X}_i) = - \sum_{\mathbf{val} \in \phi_{\mathbf{X}_i}} (P(\mathbf{X}_i = \mathbf{val}) * \log_2(P(\mathbf{X}_i = \mathbf{val})))$$

$$H(\mathbf{y}) = - \sum_{\mathbf{val} \in \phi_{\mathbf{y}}} (P(\mathbf{y} = \mathbf{val}) * \log_2(P(\mathbf{y} = \mathbf{val})))$$

and the Joint Entropy $H(\mathbf{X}_i, \mathbf{y})$ is defined by:

$$H(\mathbf{X}_i, \mathbf{y}) = - \sum_{\phi_{\mathbf{X}_i}} \left(\sum_{\phi_{\mathbf{y}}} (p_{ij} * \log_2(p_{ij})) \right), \quad p_{ij} = P(\mathbf{X}_i = \{\phi_{\mathbf{X}_i}\}_i, \mathbf{y} = \{\phi_{\mathbf{y}}\}_j)$$

where $\phi_{\mathbf{X}_i}, \phi_{\mathbf{y}}$ are sets of possible distinct values for feature $\mathbf{X}_i$ and target variable $\mathbf{y}$. Note that, the subscript i in $\mathbf{X}_i$ refers to the i^{th} feature of $\mathbf{X}$, whereas i , below the curly brackets, in $\{\phi_{\mathbf{X}_i}\}_i$ refers to the i^{th} element of the set $\phi_{\mathbf{X}_i}$.

Nonetheless, we made some modifications to suit the resource constraints and align with the objectives of this report. The primary changes were made to the hyperparameters for the Evolutionary Algorithm, i.e. `population = 10` and `generation = 25`, and we excluded the imbalance objective $o_{\text{imb}}(\cdot, \cdot)$ as it is impractical for tabular regression tasks. The modified version can be seen in `modified_mfs_split.py` in the GitHub repo provided.

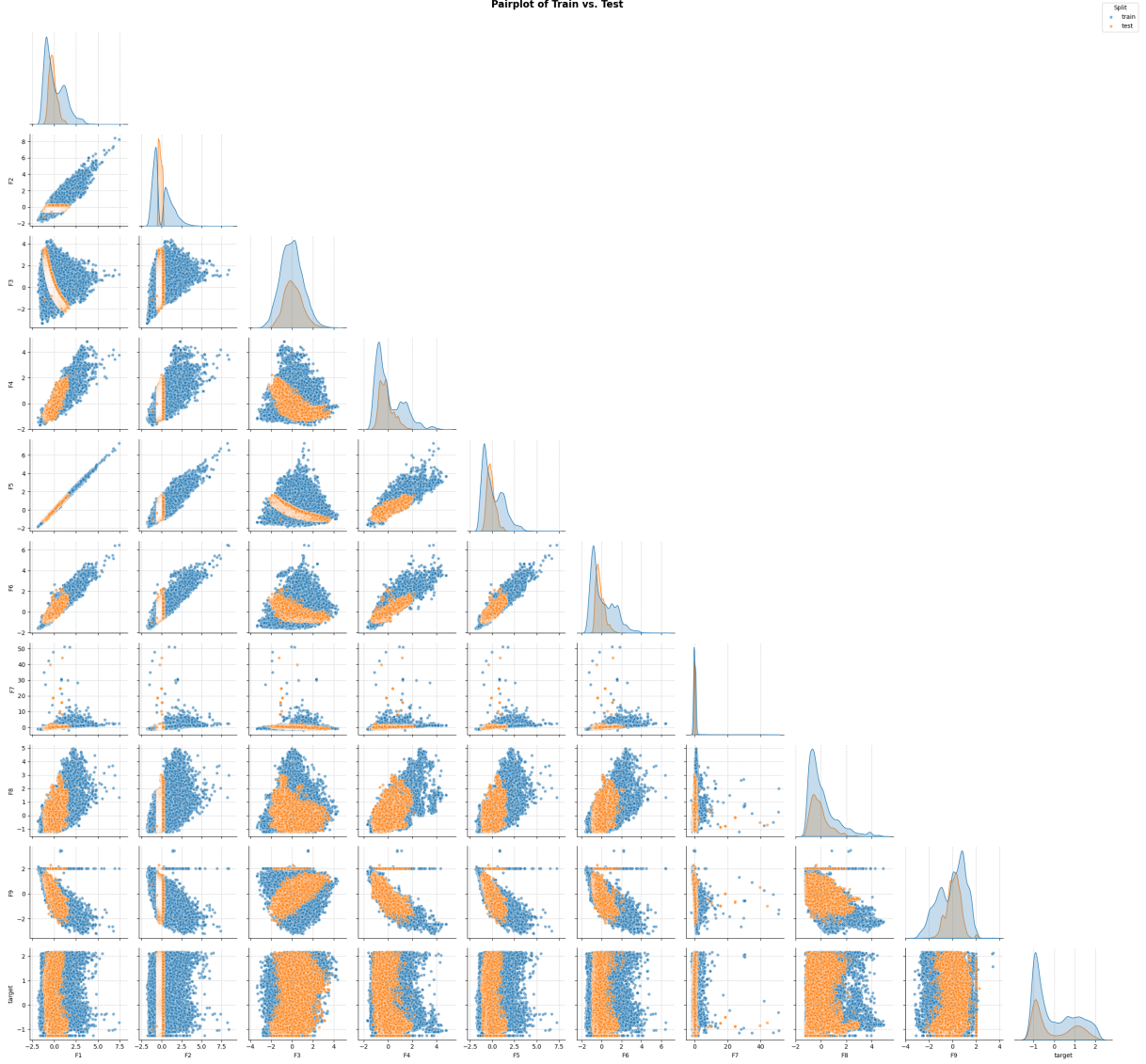


Figure 5: Pairplot of Metafeature_based_Split train_1/test_1

4.2.3 Geometry-based strategy

In this group of strategies, we utilise the **Definition 7. Interpolation vs. Extrapolation with Convex Hull** to construct train/test pairs. However, we do not explicitly apply the idea of Convex Hull. We are inspired from the geometric perspective of this definition, and thereby, proposed 5 different geometry-based approaches, i.e. Single Hyperball, Multiple Hyperballs, K-Means Hyperballs, Single Slab and Semi-infinite Slab. In the context of this report, we will be working on Euclidean space.

1. Single Hyperball

In this approach, we utilized the **Euclidean distance** of data points in the feature space $\mathcal{X}$ to split the given dataset $\mathbf{X}$. Let us define the metric used to measure the distance between two points residing in the Euclidean space.

Definition 8. Euclidean distance/metric [38]

$$\forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^n : d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|_2 = \sqrt{\sum_{i=1}^n (\mathbf{x}_i - \mathbf{y}_i)^2}$$

Now, our approach relies on the idea of the **r-Ball**. To generalize this concept, we call it **Hyperball**, similar to plane and hyperplane, which can be called in high dimensional space.

Definition 9. A Hyperball [39]

A Hyperball, or ball $\mathbf{B}_r(\mathbf{w})$ of radius r about $\mathbf{w} \in \mathbb{R}^n$ is defined as:

$$\mathbf{B}_r(\mathbf{w}) \stackrel{\text{def}}{=} \{\mathbf{x} \in \mathbb{R}^n \mid d(\mathbf{x}, \mathbf{w}) \leq r\}$$

where $d(\mathbf{x}, \mathbf{w})$ is defined by Definition 8. Indeed, it is a convex set [40].

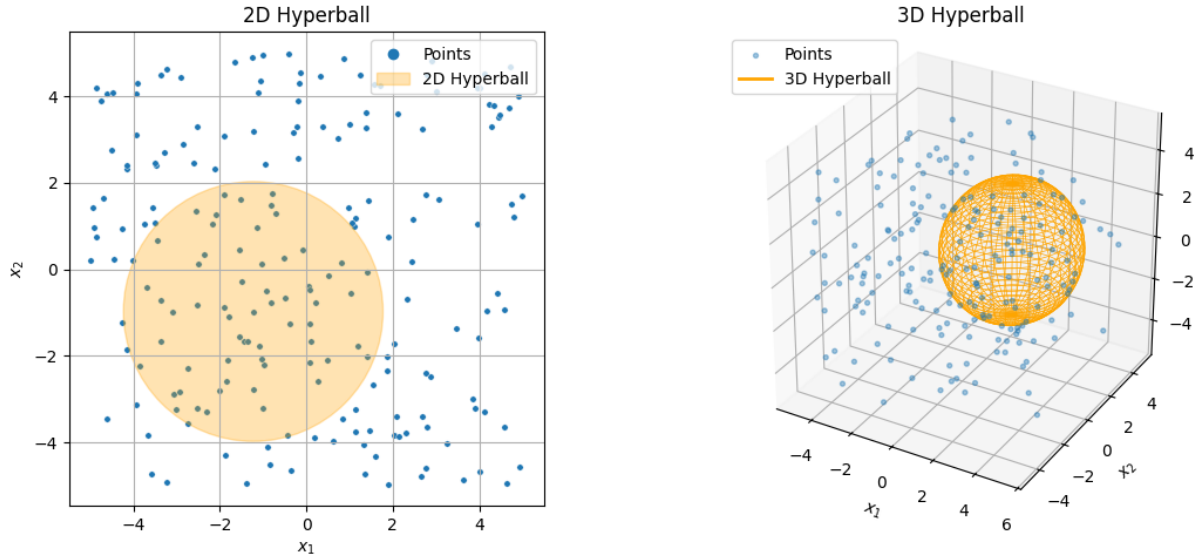


Figure 6: Hyperballs Visualization

Next, we introduce the algorithm that relies on this definition to split the datasets as following. The code below does not resemble the source code; still, it covers the core idea of the algorithm.

Algorithm 2: Single Hyperball Split (Single_Hyperball)

Require: $\mathbf{X} \in \mathbb{R}^{n \times d}$ (features), $\mathbf{y} \in \mathbb{R}^n$ (targets), train ratio $\alpha \in (0, 1)$

Ensure: A train/test split saved as `train.parquet` and `test.parquet`

- 1: $u \leftarrow \text{RandInt}(1, n)$
- 2: $\mathbf{c} \leftarrow \mathbf{x}_u \in \mathbf{X}$ ▷ Pick a random center among data points
- 3: $I_{\text{train}} \leftarrow \text{BALLSELECTION}(\mathbf{X}, \mathbf{c}, \alpha)$ ▷ Select indices inside a hyperball of size $\approx \alpha \times n$
- 4: $I_{\text{test}} \leftarrow \{1, \dots, n\} \setminus I_{\text{train}}$
- 5: $\mathbf{X}_{\text{train}} \leftarrow \mathbf{X}[I_{\text{train}}]; \mathbf{y}_{\text{train}} \leftarrow \mathbf{y}[I_{\text{train}}]$
- 6: $\mathbf{X}_{\text{test}} \leftarrow \mathbf{X}[I_{\text{test}}]; \mathbf{y}_{\text{test}} \leftarrow \mathbf{y}[I_{\text{test}}]$
- 7: $\mathcal{D}_{\text{train}} \leftarrow \text{CONCATCOLUMNS}(\mathbf{X}_{\text{train}}, \mathbf{y}_{\text{train}})$
- 8: $\mathcal{D}_{\text{test}} \leftarrow \text{CONCATCOLUMNS}(\mathbf{X}_{\text{test}}, \mathbf{y}_{\text{test}})$
- 9: Save $\mathcal{D}_{\text{train}}$ as `train.parquet`
- 10: Save $\mathcal{D}_{\text{test}}$ as `test.parquet`

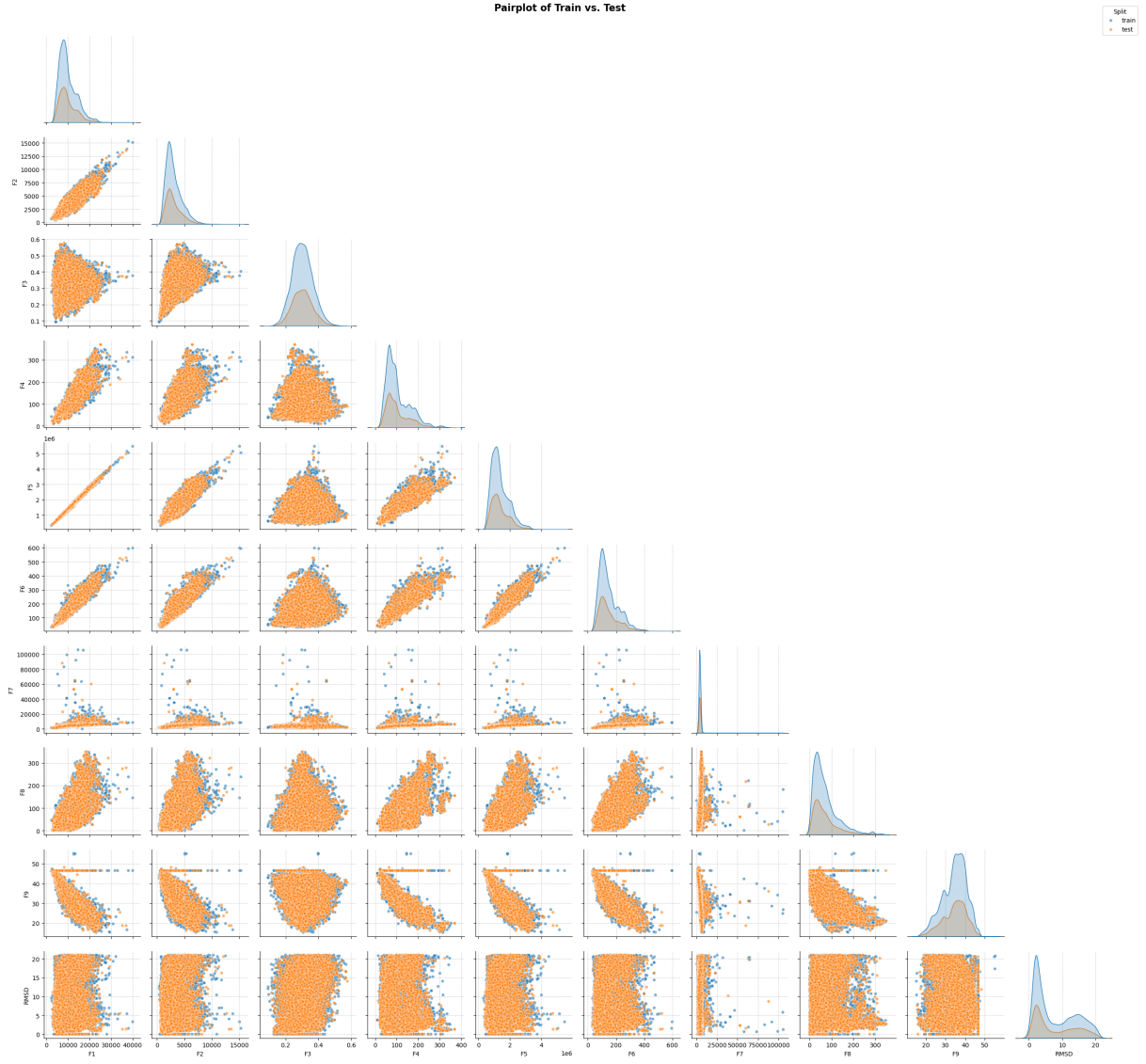


Figure 7: Pair plot of Single_Hyperball Split train_0/test_0

2. Multiple Hyperballs

In this approach, we attempt to split the feature space of the given dataset by using multiple Hyperballs in lieu of a single Hyperball. The technique can be demonstrated as follows.

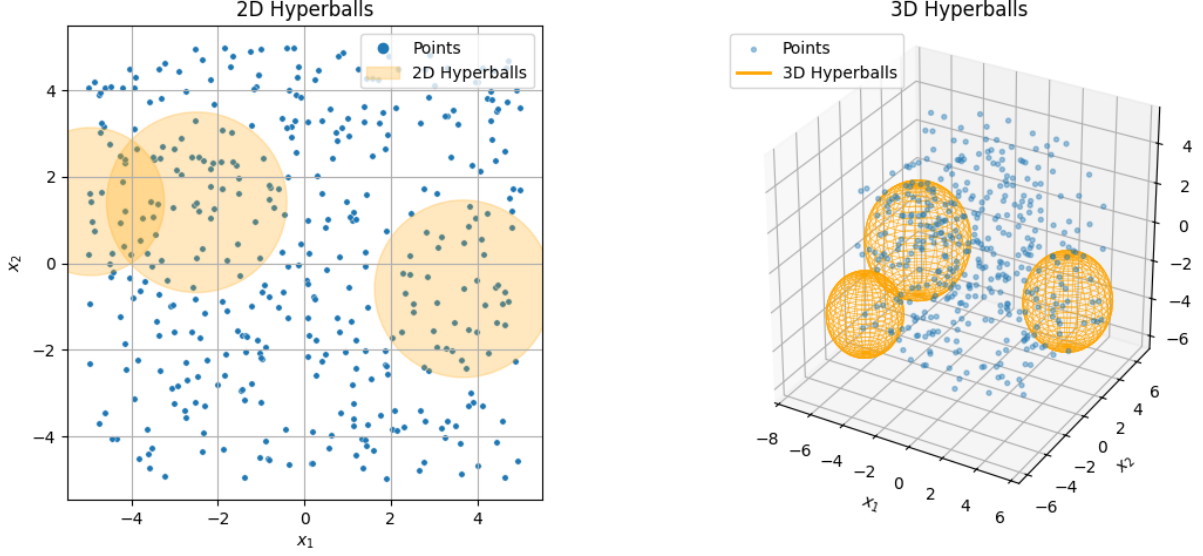


Figure 8: Multiple Hyperballs Visualization

The pseudocode is provided below, though it does not resemble the source code, it covers the primary idea of the implementation. In this report, we chose the number of Hyperballs to be 5 as default.

Algorithm 3: Multiple Hyperball Selection (Multiple_Hyperballs)

Require: $\mathbf{X} \in \mathbb{R}^{n \times d}$, $\mathbf{y} \in \mathbb{R}^n$, test ratio $\tau \in (0, 1)$, number of balls B

Ensure: A train/test split saved as `train.parquet` and `test.parquet`

```

1:  $\mathbf{X}^{(avail)} \leftarrow \mathbf{X}$ ;  $I_{test} \leftarrow \emptyset$   $\triangleright$  Work on a copy to avoid modifying the original set
2:  $(\tau_1, \tau_2, \dots, \tau_B) \leftarrow \text{RANDOMSUMS}(\tau, B)$   $\triangleright$  Split total test ratio into random parts
3: for all  $\tau_b \in (\tau_1, \dots, \tau_B)$  do
4:    $\mathbf{c} \leftarrow \text{SAMPLEONE}(\mathbf{X}^{(avail)})$   $\triangleright$  Pick a random center from available points
5:    $S \leftarrow \text{BALLSELECTION}(\mathbf{X}^{(avail)}, \mathbf{c}, \tau_b)$   $\triangleright$  Select points within a ball occupying
      about  $\tau_b$  of the dataset
6:    $S \leftarrow S \setminus I_{test}$   $\triangleright$  Avoid overlap with previously selected points
7:    $\mathbf{X}^{(avail)} \leftarrow \mathbf{X}^{(avail)} \setminus \mathbf{X}^{(avail)}[S]$   $\triangleright$  Remove selected points to diversify later centers
8:    $I_{test} \leftarrow I_{test} \cup S$   $\triangleright$  Accumulate selected test indices
9: end for
10:  $I_{train} \leftarrow \{1, \dots, n\} \setminus I_{test}$ 
11:  $\mathbf{X}_{test} \leftarrow \mathbf{X}[I_{test}]$ ;  $\mathbf{y}_{test} \leftarrow \mathbf{y}[I_{test}]$ 
12:  $\mathbf{X}_{train} \leftarrow \mathbf{X}[I_{train}]$ ;  $\mathbf{y}_{train} \leftarrow \mathbf{y}[I_{train}]$ 
13:  $\mathcal{D}_{train} \leftarrow \text{CONCATCOLUMNS}(\mathbf{X}_{train}, \mathbf{y}_{train})$ 
14:  $\mathcal{D}_{test} \leftarrow \text{CONCATCOLUMNS}(\mathbf{X}_{test}, \mathbf{y}_{test})$ 
15: Save  $\mathcal{D}_{train}$  as train.parquet
16: Save  $\mathcal{D}_{test}$  as test.parquet

```

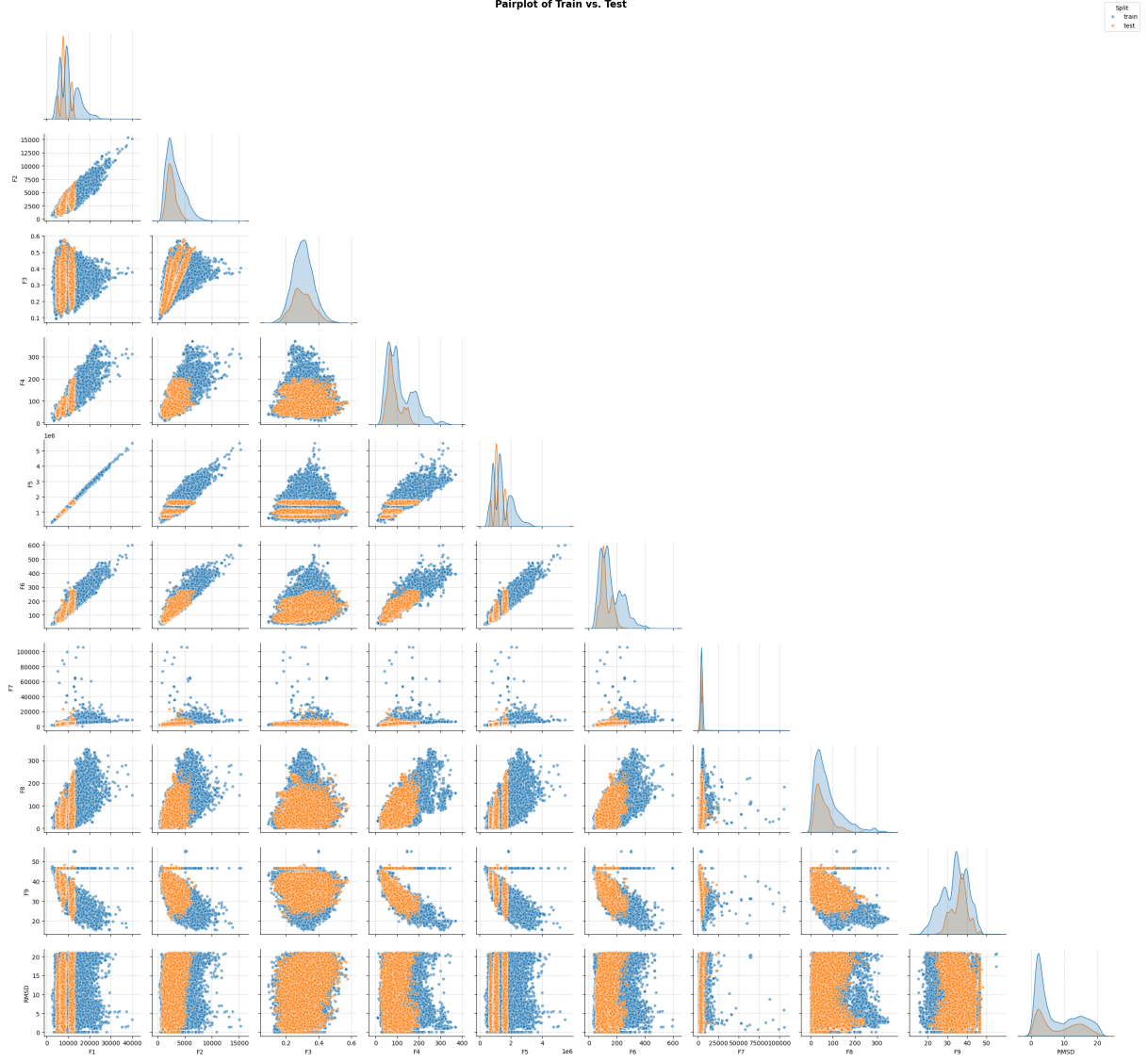


Figure 9: Pair plot of Multiple_Hyperballs Split train_0/test_0

3. K-Means Hyperballs

Developing from the previous approach, i.e. Multiple Hyperballs, we now select the centroids of the hyperballs more wisely by the utilization of the **K-Means Clustering method** ([41], [42]) implemented by Python scikit-learn library [43]. We collect the points assigned to each centroid until it reaches the expected train/test ratio. Nonetheless, this does not guarantee the selected points to be in the Hyperball about the centroids, one can either choose to follow this, or to apply the previous algorithm, i.e. Algorithm 3 with appropriate ratios for each Hyperball to construct the train/test pair, they are not significantly different. In this report, we chose the number of clusters to be 5 and the default algorithm is **Lloyd's algorithm** [42]. Since we leverage the implementation of scikit-learn library [43], we do not introduce to pseudocode of this method. Alternatively, we propose the problem formulation and objective of the solution.

Problem statement: Given a set of n data points in the feature space $\mathbf{X}$ and no label space $\mathbf{y}$, which is known as an **unsupervised problem** [42]. Group the data points into different clusters so that each cluster shares similar characteristics among the data points

residing in the same cluster. The problem can be mathmematically formulized as below.

Problem formulation [41]: Given a dataset $\mathbf{X} = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n \mid \mathbf{x}_i \in \mathbb{R}^m\}$ of n points. Partition $\mathbf{X}$ into K clusters, denoted as $\mathcal{C} = \{C_1, C_2, \dots, C_K\}$, subject to:

$$\begin{aligned} C_i \cap C_j &= \emptyset, \forall C_i, C_j \in \mathcal{C} \\ \bigcup_{i=1}^K C_i &= \mathbf{X} \\ \arg \min_{\mathcal{C}} \sum_{k=1}^K \sum_{\mathbf{x}_i \in C_k} \|\mathbf{x}_i - \boldsymbol{\mu}_k\|_2^2 \end{aligned}$$

where:

- $\|\cdot\|_2^2$ is the squared Euclidean distance.
- $\boldsymbol{\mu}_k = \frac{\sum_{\mathbf{x}_i \in C_k} \mathbf{x}_i}{|C_k|}$ is the centroid of the cluster C_k .

Lloyd's algorithm ([42], [41]):

Step 1. Clusters' centroids random initialization, i.e. $\boldsymbol{\mu} = \{\boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \dots, \boldsymbol{\mu}_K\}$

Step 2. Assign the points $\mathbf{x}_i \in \mathbf{X}$ to their nearest centroid. More specifically,

$$z(\mathbf{x}_i) = \arg \min_{k \in \overline{1, K}} \|\mathbf{x}_i - \boldsymbol{\mu}_k\|_2^2 \Rightarrow C_z = C_z \cup \mathbf{x}_i$$

Step 3. Centroids (re-)calculation

$$k \in \overline{1, K} : \quad \boldsymbol{\mu}_k = \frac{\sum_{\mathbf{x}_i \in C_k} \mathbf{x}_i}{|C_k|}$$

then empty the sets $C_1, \dots, C_K = \emptyset$

Repeat Step 2 and Step 3 until convergence, i.e. the clusters' centroids remain unchanged.

Algorithm 4: **K-Means Hyperballs (KMeans_Hyperballs)**

Require: $\mathbf{X} \in \mathbb{R}^{n \times d}$, $\mathbf{y} \in \mathbb{R}^n$, test ratio $\tau \in (0, 1)$, number of clusters K

Ensure: A train/test split saved as `train.parquet` and `test.parquet`

- 1: $kmeans \leftarrow \text{KMEANS}(n_clusters = K)$ ▷ KMEANS($\cdot$) is adopted from sklearn library [43]
- 2: $\text{FIT}(kmeans, \mathbf{X})$
- 3: $centroids \leftarrow kmeans.cluster_centers_$
- 4: $labels \leftarrow kmeans.labels_$
- 5: $total \leftarrow n$
- 6: $clusters \leftarrow \{0, 1, \dots, K - 1\}$
- 7: $I_{\text{test}} \leftarrow \emptyset$ ▷ indices selected for test set
- 8: **while** $\frac{|I_{\text{test}}|}{total} < \tau$ **and** $clusters \neq \emptyset$ **do**
- 9: $cl \leftarrow \text{RANDOMCHOICE}(clusters)$
- 10: $C \leftarrow \{i \in \overline{1, n} : labels[i] = cl\}$ ▷ indices of chosen cluster
- 11: **if** $\frac{|I_{\text{test}}| + |C|}{total} < \tau$ **then**
- 12: $I_{\text{test}} \leftarrow I_{\text{test}} \cup C$


```

13:   else
14:        $num \leftarrow \lfloor \tau \cdot total - |I_{test}| \rfloor$ 
15:        $I_{test} \leftarrow I_{test} \cup \text{FIRSTK}(C, num)$  ▷ take first  $num$  indices
16:   end if
17:    $clusters \leftarrow clusters \setminus \{cl\}$  ▷ prevent re-selecting the same cluster
18: end while
19:  $I_{train} \leftarrow \{1, \dots, n\} \setminus I_{test}$ 
20:  $\mathbf{X}_{train} \leftarrow \mathbf{X}[I_{train}]; \mathbf{y}_{train} \leftarrow \mathbf{y}[I_{train}]$ 
21:  $\mathbf{X}_{test} \leftarrow \mathbf{X}[I_{test}]; \mathbf{y}_{test} \leftarrow \mathbf{y}[I_{test}]$ 
22:  $\mathcal{D}_{train} \leftarrow \text{CONCATCOLUMNS}(\mathbf{X}_{train}, \mathbf{y}_{train})$ 
23:  $\mathcal{D}_{test} \leftarrow \text{CONCATCOLUMNS}(\mathbf{X}_{test}, \mathbf{y}_{test})$ 
24: Save  $\mathcal{D}_{train}$  as train.parquet
25: Save  $\mathcal{D}_{test}$  as test.parquet

```

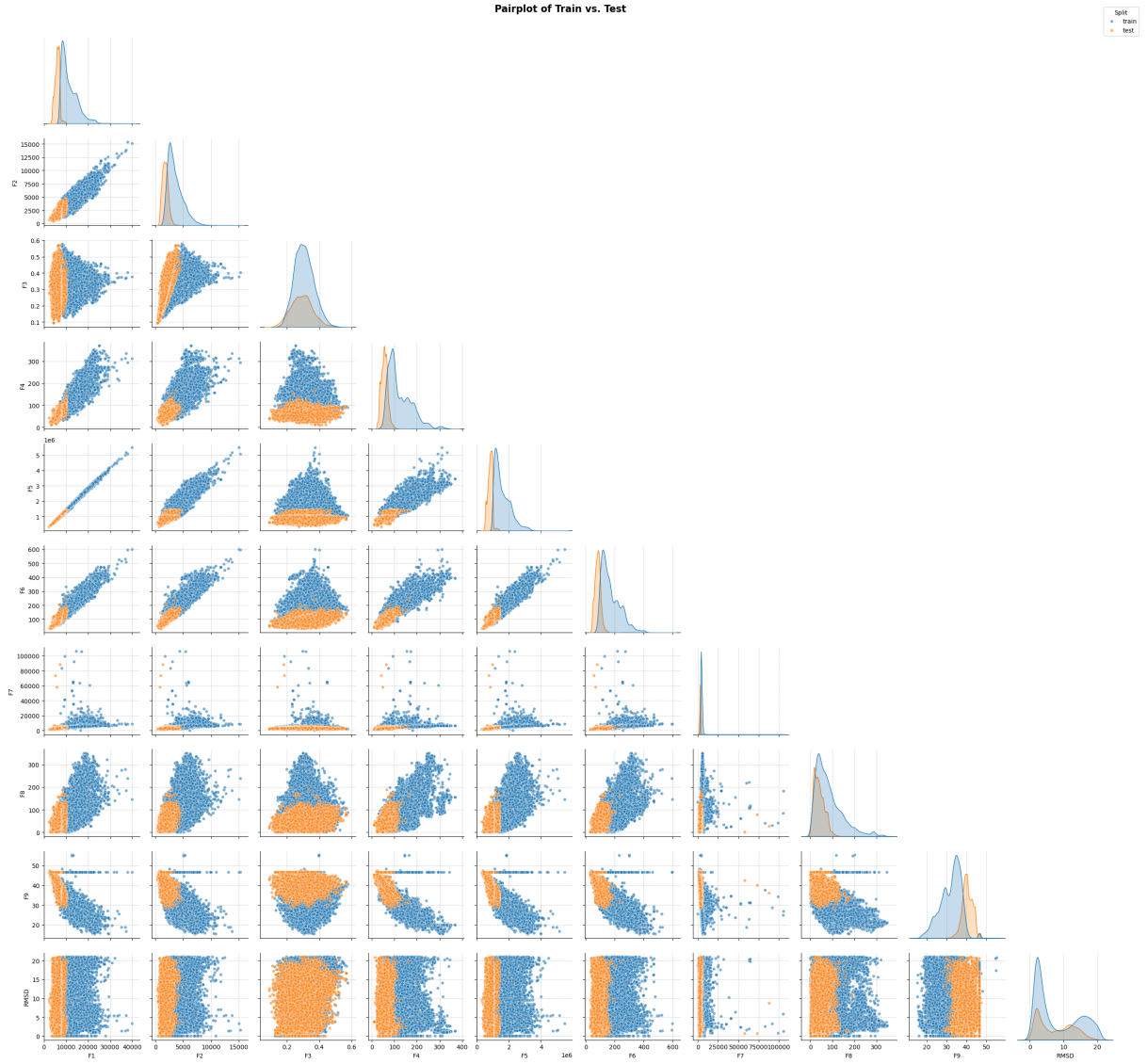


Figure 10: Pair plot of KMeans_Hyperballs Split train_0/test_0

In the last two splitting techniques, we are inspired by the geometric perspectives of **single slab** and **semi-infinite slab** from Partial Differential Equations [44], not their

calculus meaning. Thus, we utilize the notions of planes, vectors and distance to construct these two kinds of slab.

4. Single Slab

Firstly, let us formally review certain useful notions.

Definition 10. A Hyperplane [40]

A Hyperplane P is a set given by:

$$P \stackrel{\text{def}}{=} \{\mathbf{x} \in \mathbb{R}^n \mid \mathbf{n}^\top \mathbf{x} = b\}$$

where $\mathbf{n} \in \mathbb{R}^n \setminus \{\mathbf{0}\}$ is called the normal vector, which is normal/orthogonal to P , i.e. $\mathbf{n} \perp P$, and $b \in \mathbb{R}$.

Said differently, P is defined by a normal vector $\mathbf{n}$ and a point $\mathbf{x}_0 \in \mathbb{R}^n$. The above set turns out to be:

$$P \stackrel{\text{def}}{=} \{\mathbf{x} \in \mathbb{R}^n \mid \mathbf{n}^\top (\mathbf{x} - \mathbf{x}_0) = 0\}$$

then $b = \mathbf{n}^\top \mathbf{x}_0$.

It is imperative to know how to measure the distance of a point $\mathbf{x}$ to a given hyperplane P , as it determines whether the points belong to train or test sets.

Definition 11. (Shortest) Distance from a point to a Hyperplane [39]

Given a point $\mathbf{w} \in \mathbb{R}^n$ and a hyperplane $P = \{\mathbf{x} \mid \mathbf{n}^\top \mathbf{x} = b\} \subset \mathbb{R}^n$. The shortest distance from $\mathbf{w}$ to P is defined as:

$$d(\mathbf{x}, P) \stackrel{\text{def}}{=} \frac{|\mathbf{n}^\top \mathbf{x} - b|}{\|\mathbf{n}\|_2}$$

From the above definitions, we propose a useful notation of δ -slab, which is leveraged to partition the datasets.

Definition 12. A δ -slab

Given a hyperplane $P = \{\mathbf{x} \mid \mathbf{n}^\top \mathbf{x} = b\} \subset \mathbb{R}^n$. A δ -slab defined by hyperplane P , is given by:

$$Slab_\delta(P) \stackrel{\text{def}}{=} \{\mathbf{x} \in \mathbb{R}^n \mid d(\mathbf{x}, P) < \delta\} \subset \mathbb{R}^n, \quad \text{for } \delta \in \mathbb{R}$$

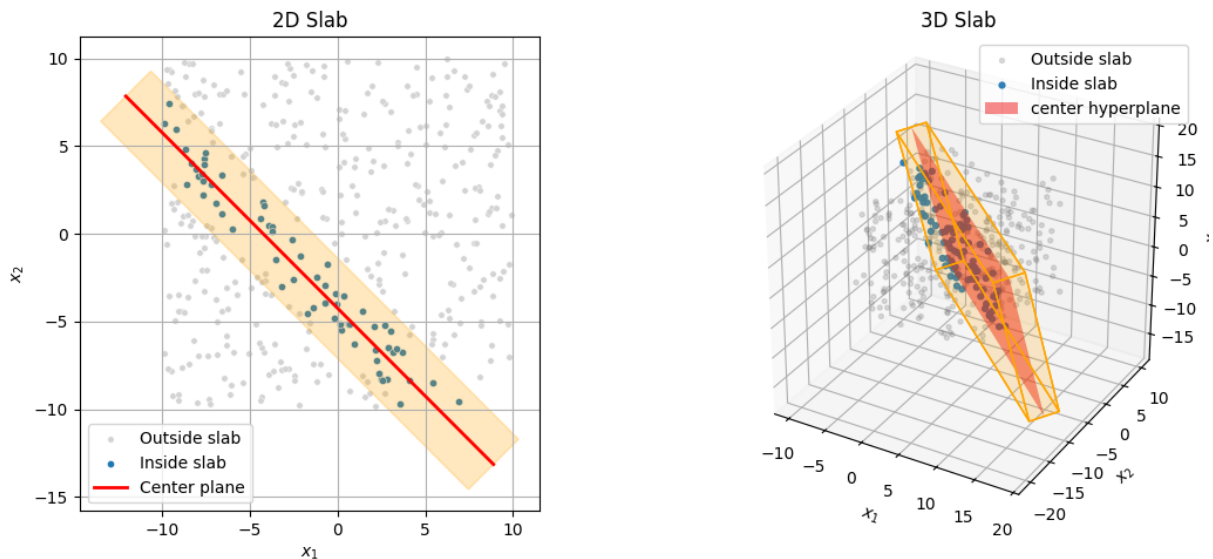


Figure 11: δ -slab Visualization

Next, we propose the following algorithm that follows the concept of δ -slab discussed above to split the given datasets into train/test pairs.

Algorithm 5: **Single Slab (Single_Slab)**

Require: $\mathbf{X} \in \mathbb{R}^{n \times d}$, $\mathbf{y} \in \mathbb{R}^n$, train ratio $\alpha \in (0, 1)$

Ensure: A train/test split saved as `train.parquet` and `test.parquet`

```

1:  $\varepsilon \leftarrow 10^{-2}$ 
2:  $(\mathbf{n}, b, \mathbf{p}_0) \leftarrow \text{CONSTRUCTHYPERPLANE}(\mathbf{X})$ 
    $\triangleright$  Binary search for  $\delta$  so that  $\approx \alpha \times n$  points lie within the  $\delta$ -slab
3:  $lo \leftarrow 0$ 
4:  $high \leftarrow \max_{i \in \overline{1, n}} \|\mathbf{X}_i - \mathbf{p}\|_2$ 
5: while  $lo < high$  do
6:
7:    $\delta \leftarrow lo + \frac{high - lo}{2}$ 
8:
9:    $I_{\text{train}} \leftarrow \left\{ i \in \overline{1, n} : \frac{|\mathbf{n}^\top \mathbf{X}_i - b|}{\|\mathbf{n}\|_2} < \delta \right\}$   $\triangleright$  indices within the  $\delta$ -slab
10:  if  $\frac{|I_{\text{train}}|}{n} < \alpha$  then
11:     $lo \leftarrow \delta + \varepsilon$ 
12:  else
13:     $high \leftarrow \delta - \varepsilon$ 
14:  end if
15: end while
16:  $I_{\text{test}} \leftarrow \{1, \dots, n\} \setminus I_{\text{train}}$ 
17:  $\mathbf{X}_{\text{train}} \leftarrow \mathbf{X}[I_{\text{train}}]$ ;  $\mathbf{y}_{\text{train}} \leftarrow \mathbf{y}[I_{\text{train}}]$ 
18:  $\mathbf{X}_{\text{test}} \leftarrow \mathbf{X}[I_{\text{test}}]$ ;  $\mathbf{y}_{\text{test}} \leftarrow \mathbf{y}[I_{\text{test}}]$ 
19:  $\mathcal{D}_{\text{train}} \leftarrow \text{CONCATCOLUMNS}(\mathbf{X}_{\text{train}}, \mathbf{y}_{\text{train}})$ 
20:  $\mathcal{D}_{\text{test}} \leftarrow \text{CONCATCOLUMNS}(\mathbf{X}_{\text{test}}, \mathbf{y}_{\text{test}})$ 
21: Save  $\mathcal{D}_{\text{train}}$  as train.parquet
22: Save  $\mathcal{D}_{\text{test}}$  as test.parquet

```

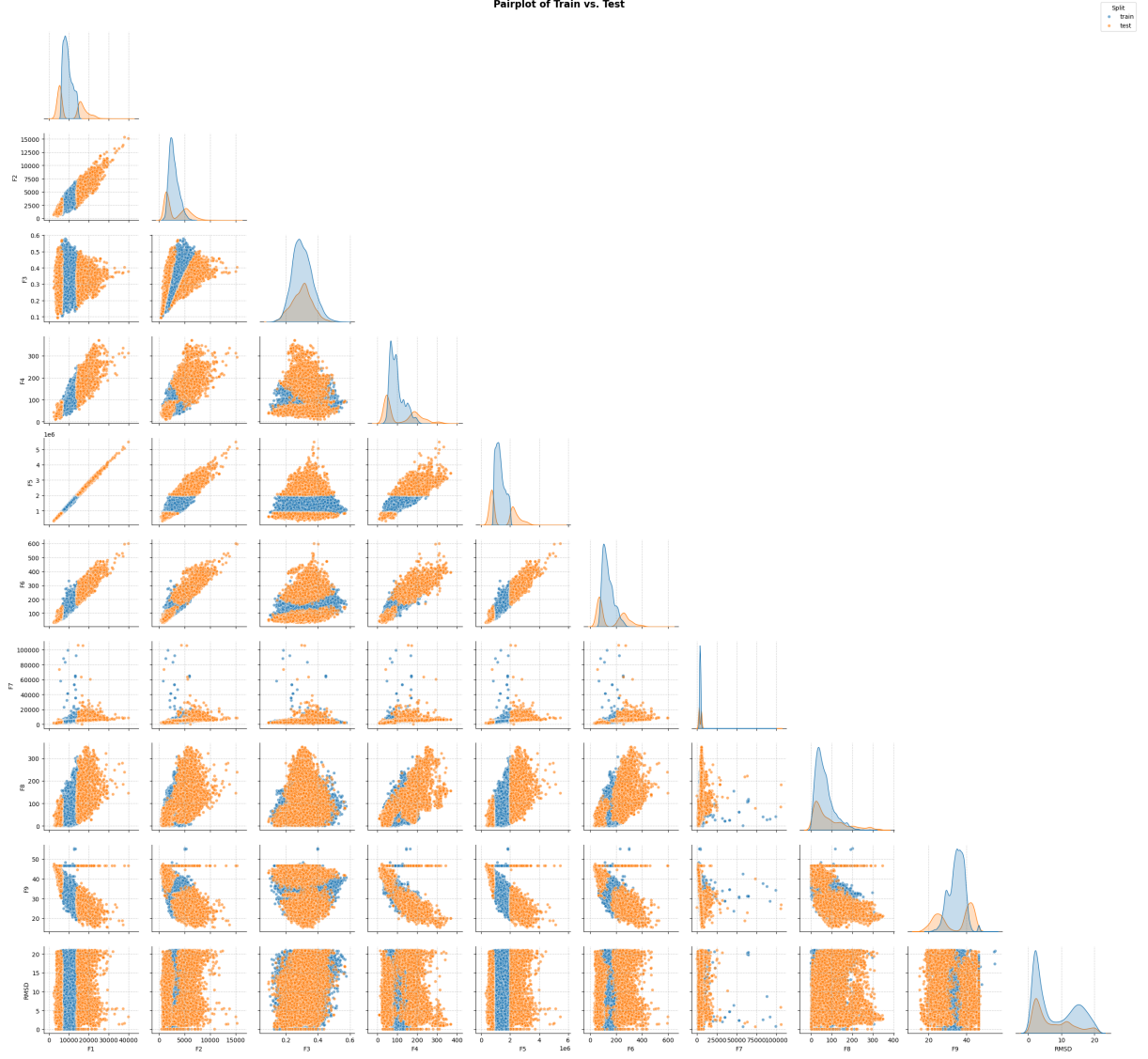


Figure 12: Pair plot of Single_Slab Split train_0/test_0

5. Semi-infinite Slab

In this final approach, we utilise the hyperplane P defined in **Definition 10**. **A Hyperplane** to split the given datasets. [40] A hyperplane $P \subset \mathbb{R}^n$ partitions the space into two halves, i.e. a (closed) halfspace $\mathcal{S}_1 = \{\mathbf{x} \mid \mathbf{n}^\top \mathbf{x} \leq b\}$ and an (open) halfspace $\mathcal{S}_2 = \{\mathbf{x} \mid \mathbf{n}^\top \mathbf{x} > b\}$, which hold $\mathbb{R}^n = \mathcal{S}_1 \cup \mathcal{S}_2$ and $\mathcal{S}_1 \cap \mathcal{S}_2 = \emptyset$. In our testbench, we use the (closed) halfspace to construct the test set and the remaining will be allocated to the train set. More formally:

Definition 13. A semi-infinite slab

Given a hyperplane $P = \{\mathbf{x} \mid \mathbf{n}^\top \mathbf{x} = b\} \subset \mathbb{R}^n$. **A semi-infinite slab** defined by hyperplane P , is given by:

$$Slab_\infty(P) \stackrel{\text{def}}{=} \{\mathbf{x} \in \mathbb{R}^n \mid \mathbf{n}^\top \mathbf{x} < b\}$$

Indeed, it is a convex set [40].

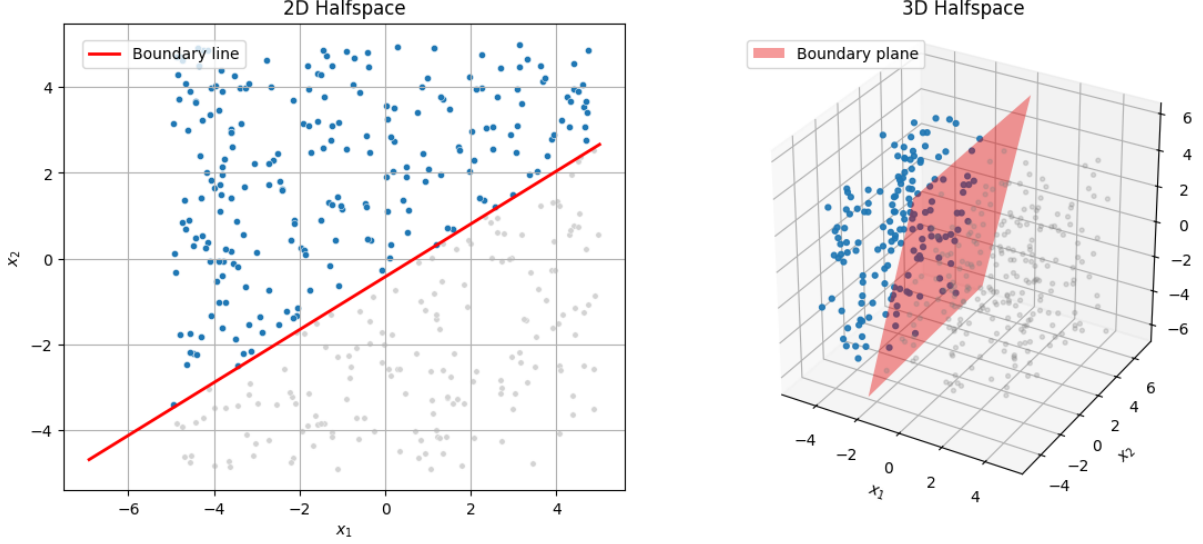


Figure 13: Semi-infinite slab Visualization

Next, we propose the following algorithm that follows the concept of semi-infinite slab discussed above to split the given datasets into train/test pairs.

Algorithm 6: **Semi-Infinite Slab (Semi_Infinite_Slab)**

Require: $\mathbf{X} \in \mathbb{R}^{n \times d}$, $\mathbf{y} \in \mathbb{R}^n$, test ratio $\tau \in (0, 1)$

Ensure: A train/test split saved as `train.parquet` and `test.parquet`

- 1: $\varepsilon \leftarrow 0.01$; $cur \leftarrow 0$
- 2: $(\mathbf{n}, \mathbf{lo}, \mathbf{hi}) \leftarrow \text{FINDBOUNDS}(\mathbf{X}, \tau)$ $\triangleright$ $\mathbf{n}$ is a random normal; $\mathbf{lo}, \mathbf{hi}$ are bounding points for binary search for appropriate points to construct the hyperplane
- 3: **while** $cur < \tau$ **or** $cur > \tau + \varepsilon$ **do**
- 4: $\mathbf{p} \leftarrow \mathbf{lo} + \frac{\mathbf{hi} - \mathbf{lo}}{2}$
- 5: $b \leftarrow \mathbf{n}^\top \mathbf{p}$
- 6: $I_{\text{test}} \leftarrow \text{DATAONESIDE}(\mathbf{X}, \mathbf{n}, b)$
- 7: $cur \leftarrow \frac{|I_{\text{test}}|}{n}$
- 8: **if** $cur < \tau$ **then**
- 9: $\mathbf{lo} \leftarrow \mathbf{p} + \varepsilon$ $\triangleright$ shift lower bound upward
- 10: **else**
- 11: $\mathbf{hi} \leftarrow \mathbf{p} - \varepsilon$ $\triangleright$ shift upper bound downward
- 12: **end if**
- 13: **end while**
- 14: $I_{\text{train}} \leftarrow \{1, \dots, n\} \setminus I_{\text{test}}$
- 15: $\mathbf{X}_{\text{train}} \leftarrow \mathbf{X}[I_{\text{train}}]$; $\mathbf{y}_{\text{train}} \leftarrow \mathbf{y}[I_{\text{train}}]$
- 16: $\mathbf{X}_{\text{test}} \leftarrow \mathbf{X}[I_{\text{test}}]$; $\mathbf{y}_{\text{test}} \leftarrow \mathbf{y}[I_{\text{test}}]$
- 17: $\mathcal{D}_{\text{train}} \leftarrow \text{CONCATCOLUMNS}(\mathbf{X}_{\text{train}}, \mathbf{y}_{\text{train}})$
- 18: $\mathcal{D}_{\text{test}} \leftarrow \text{CONCATCOLUMNS}(\mathbf{X}_{\text{test}}, \mathbf{y}_{\text{test}})$
- 19: Save $\mathcal{D}_{\text{train}}$ as `train.parquet`
- 20: Save $\mathcal{D}_{\text{test}}$ as `test.parquet`

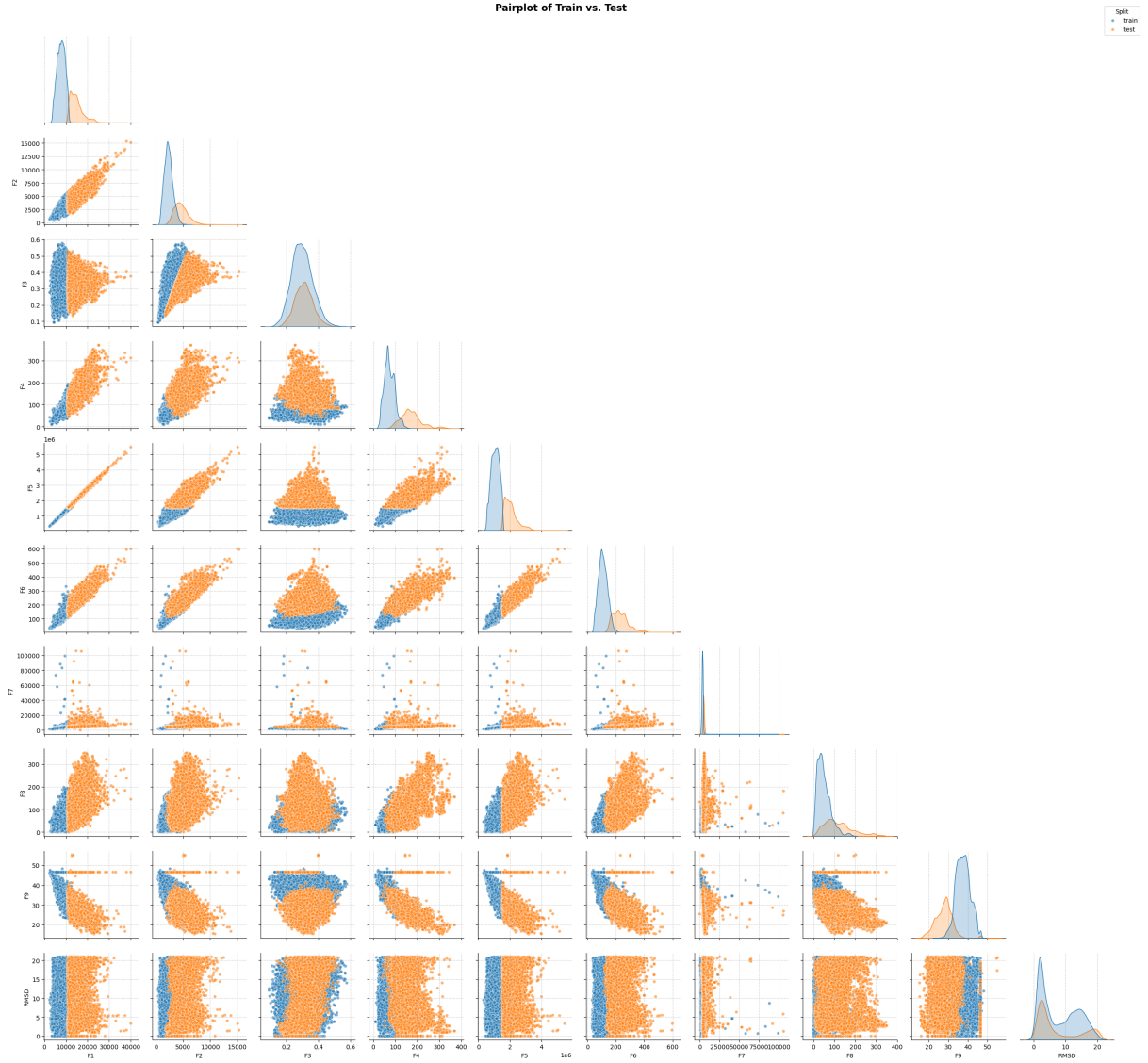


Figure 14: Pair plot of Semi-infinite_Slab Split train_0/test_0

4.3 Models

5 Experimental setup

5.1 Evaluation metrics

5.2 Hyperparameters selection methods

For each training data, the model selection method is re-run to find the most optimal hyperparameter setting w.r.t the dataset.

6 Results

7 Analysis

8 Conclusion and Discussion

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Appendices

A Notation and Abbreviation Reference List

Abbreviation

Acronyms	Meaning
tr	train
te	test
emp	empirical
vs.	versus
iid	independent and identically distributed
ID	In-Distribution
OOD	Out-of-Distribution
MSE	Mean Squared Error
MAE	Mean Absolute Error
RMSE	Root Mean Squared Error
$\mathbf{R}^2$	Coefficient of Determination

Notation

Symbol	Meaning
$\mathcal{R}(\cdot)$	Risk
$\mathcal{X}$	Input/Feature space or Domain
$\mathcal{Y}$	Output/Label space or Codomain
$ \mathcal{X} $	Size, or number of elements of set $\mathcal{X}$
$\mathbf{X}^{n \times m}$	Matrix of Covariates/Features of m features and n samples
$\mathbf{y}^{n \times 1}$	Target vector of n outputs
$\mathbb{E}$	Expected Value
$\mathbf{P}(\mathbf{X}, \mathbf{y})$	Joint Distribution
$\mathbf{P}(\mathbf{X}), \mathbf{P}(\mathbf{y})$	Marginal Distribution
$\mathbf{P}(\mathbf{y} \mid \mathbf{X})$	Conditional Distribution
$(\mathbf{x}_i, y_i) \sim \mathbf{P}(\cdot)$	A data point sampled from a distribution $\mathbf{P}(\cdot)$
$f(\mathbf{x}_i) \mapsto y_i$	True mapping from $\mathbf{x}_i \in \mathbf{X}$ onto $y_i \in \mathbf{y}$
$\hat{y}_i = \hat{f}(\cdot) \approx y_i$	An approximation mapping/function
$\ell(f(\mathbf{x}_i), y_i) \mapsto \mathbb{R}_{\geq 0}$	loss of a pair of input and true output $(\mathbf{x}_i, y_i)$
$\text{Conv}(\cdot)$	Convex hull
$\ \cdot\ _p = [\sum (\cdot)^p]^{\frac{1}{p}}$	ℓ_p -norm
$\overline{a, b}$	$\{a, a+1, \dots, b-1, b\}$
$\lfloor \cdot \rfloor, \lceil \cdot \rceil$	approximate the nearest smaller/larger integer

B Supportive algorithms

This section aims to providing additional functions that support the core algorithms presented within this report.

Algorithm 7: Covariate Distribution Based Selection

Require: Dataset $\mathcal{D} = (\mathbf{X}, \mathbf{y})$ with features $\{\mathbf{x}_1, \dots, \mathbf{x}_m\}$ and target $\mathbf{y}$; test ratio $\tau \in (0, 1)$; feat: selected feature to split

Ensure: *mask*: selected samples

- 1: $lo \leftarrow 0$; $hi \leftarrow 1$; $\varepsilon \leftarrow 10^{-3}$
- 2: **while** $lo < hi$ **do**
- 3: $mid \leftarrow lo + \frac{hi - lo}{2}$

```

4:    $q_{low} \leftarrow Q_{\text{feat}}(mid); q_{high} \leftarrow Q_{\text{feat}}(1 - mid)$ 
5:    $mask \leftarrow \mathbb{1}\{\text{DIGITIZE}(\mathbf{X}_{\text{feat}}; q_{low}, q_{high}) = 1\}$ 
6:   if  $\frac{\sum mask}{|\mathcal{D}|} < \tau$  then
7:      $lo \leftarrow mid + \varepsilon$ 
8:   else
9:      $hi \leftarrow mid - \varepsilon$ 
10:  end if
11: end while
12: return  $mask$ 

```

Algorithm 8: BALLSELECTION (binary search on radius)

Require: $\mathbf{X} \in \mathbb{R}^{n \times d}$, center $\mathbf{c} \in \mathbb{R}^d$, train ratio α
Ensure: $I \subseteq \{1, \dots, n\}$, indices of points inside the selected ball

```

1:  $\varepsilon \leftarrow 10^{-3}; lo \leftarrow 0$ 
2:  $high \leftarrow \max_{i \in \{1, \dots, n\}} \|\mathbf{X}_i - \mathbf{c}\|_2$ 
3: while  $lo < high$  do
4:    $mid \leftarrow lo + \frac{high - lo}{2}$ 
5:    $I \leftarrow \{i \in \overline{1, n} : \|\mathbf{X}_i - \mathbf{c}\|_2 \leq mid\}$ 
6:   if  $\frac{|I|}{n} < \alpha$  then
7:      $lo \leftarrow mid + \varepsilon$ 
8:   else
9:      $high \leftarrow mid - \varepsilon$ 
10:  end if
11: end while
12: return  $I$ 

```

Algorithm 9: RANDOMSUMS (random partition of a total fraction)

Require: Total fraction τ , number of balls B
Ensure: $(\tau_1, \dots, \tau_B)$ such that $\sum_{b=1}^B \tau_b = \tau$

```

1: if  $B = 1$  then
2:   return  $(\tau)$ 
3: end if
4: Generate  $B - 1$  i.i.d. cuts  $u_1, \dots, u_{B-1} \sim \text{Uniform}(0, \tau)$ 
5: Sort cuts to get  $u_{(1)} \leq \dots \leq u_{(B-1)}$ 
6:  $\tau_1 \leftarrow u_{(1)}$ 
7: for  $b = 2$  to  $B - 1$  do
8:    $\tau_b \leftarrow u_{(b)} - u_{(b-1)}$ 
9: end for
10:  $\tau_B \leftarrow \tau - u_{(B-1)}$ 
11: return  $(\tau_1, \dots, \tau_B)$ 

```

Algorithm 10: Hyperplane Construction (CONSTRUCTHYPERPLANE)

Require: $\mathbf{X} \in \mathbb{R}^{n \times d}$

Ensure: Normal vector $\mathbf{n}$, offset b , and a point $\mathbf{p}_0$ on the hyperplane

- 1: $u \leftarrow \text{RANDINT}(1, n)$
- 2: $\mathbf{p}_0 \leftarrow \mathbf{X}_u$
- 3: $\mathbf{n} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$
- 4: $b \leftarrow \mathbf{n}^\top \mathbf{p}_0$
- 5: **return** $(\mathbf{n}, b, \mathbf{p}_0)$

Algorithm 11: **One-Side Selection** (DATAONESIDE)

Require: $\mathbf{X} \in \mathbb{R}^{n \times d}$, normal vector $\mathbf{n}$, offset $b \in \mathbb{R}$

Ensure: Index set $I = \{i \in \overline{1, n} : \mathbf{X}_i^\top \mathbf{n} < b\}$

- 1: $I \leftarrow \{i \in \overline{1, n} : \mathbf{X}_i^\top \mathbf{n} - b < 0\}$
- 2: **return** I

Algorithm 12: **Find Bounds for Semi-Infinite Slab** (FINDBOUNDS)

Require: $\mathbf{X} \in \mathbb{R}^{n \times d}$, test ratio $\tau \in (0, 1)$

Ensure: Normal vector $\mathbf{n}$, lower/upper bounding points $(\mathbf{lo}, \mathbf{hi})$

- 1: $(\mathbf{n}, b, \mathbf{p}_0) \leftarrow \text{CONSTRUCTHYPERPLANE}(\mathbf{X})$
- 2: $dist \leftarrow \frac{|\mathbf{X}^\top \mathbf{n} - b|}{\|\mathbf{n}\|_2}$
- 3:
- 4: $mask \leftarrow (\mathbf{X}^\top \mathbf{n} - b < 0)$
- 5: $L \leftarrow \{i \in \overline{1, n} : mask_i = \text{True}\}; H \leftarrow \{i \in \overline{1, n} : mask_i = \text{False}\}$
 $\triangleright$ Pick farthest points on each side; fallback if one side is empty
- 6: **if** $L \neq \emptyset$ **and** $H \neq \emptyset$ **then**
- 7: $i_{lo} \leftarrow \arg \max_{i \in L} dist_i; \quad i_{hi} \leftarrow \arg \max_{i \in H} dist_i$
- 8: **else**
- 9: $i_{lo} \leftarrow \arg \min_{i \in \overline{1, n}} dist_i; \quad i_{hi} \leftarrow \arg \max_{i \in \overline{1, n}} dist_i$
- 10: **end if**
- 11: $\mathbf{lo} \leftarrow \mathbf{X}_{i_{lo}}; \quad \mathbf{hi} \leftarrow \mathbf{X}_{i_{hi}}$
- 12:
- 13: $b_{hi} \leftarrow \mathbf{n}^\top \mathbf{hi}$
- 14: $I_{\text{test}} \leftarrow \text{DATAONESIDE}(\mathbf{X}, \mathbf{n}, b_{hi})$
- 15: **if** $\frac{|I_{\text{test}}|}{n} < \tau$ **then**
- 16: $\text{swap}(\mathbf{lo}, \mathbf{hi})$ $\triangleright$ ensure upper bound can reach at least τ
- 17: **end if**
- 18: **return** $(\mathbf{n}, \mathbf{lo}, \mathbf{hi})$

C Hyperparameter setup

D Benchmark results